# SC2107/CE2107 Microprocessor System Design and Development

## Tutorial 6

## Serial Bus and Display

1. Discuss the common and different characteristics of SPI and I2C?

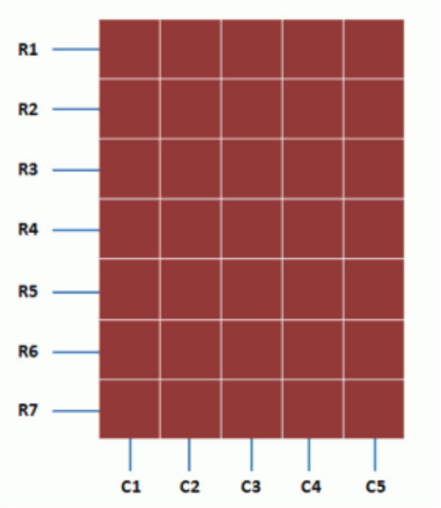
2. In the I2C protocol, in order to read from the slave device, you must first tell it which of its internal addresses you want to read from. So a read of the slave actually starts off by writing to it. Let's look at the sequence of communicating with a peripheral module.
   - Send the Start Sequence
   - Send 0xC0 (I2C address of the peripheral module with R/W set to low)
   - Send 0x01 (Internal address of the register that we want to read from)
   - Send a Start Sequence again
   - Send 0xC1 (I2C address of the peripheral module with R/W set to high)
   - Read a byte from the peripheral device
   - Send the Stop Sequence

   Illustrate the SCL and SDA lines of the I2C bus to reflect the above sequence.

3. In the I2C protocol, there could be a situation where the slave is not ready to send data.

   Why is this a problem and describe the solution in the I2C protocol to overcome this problem.

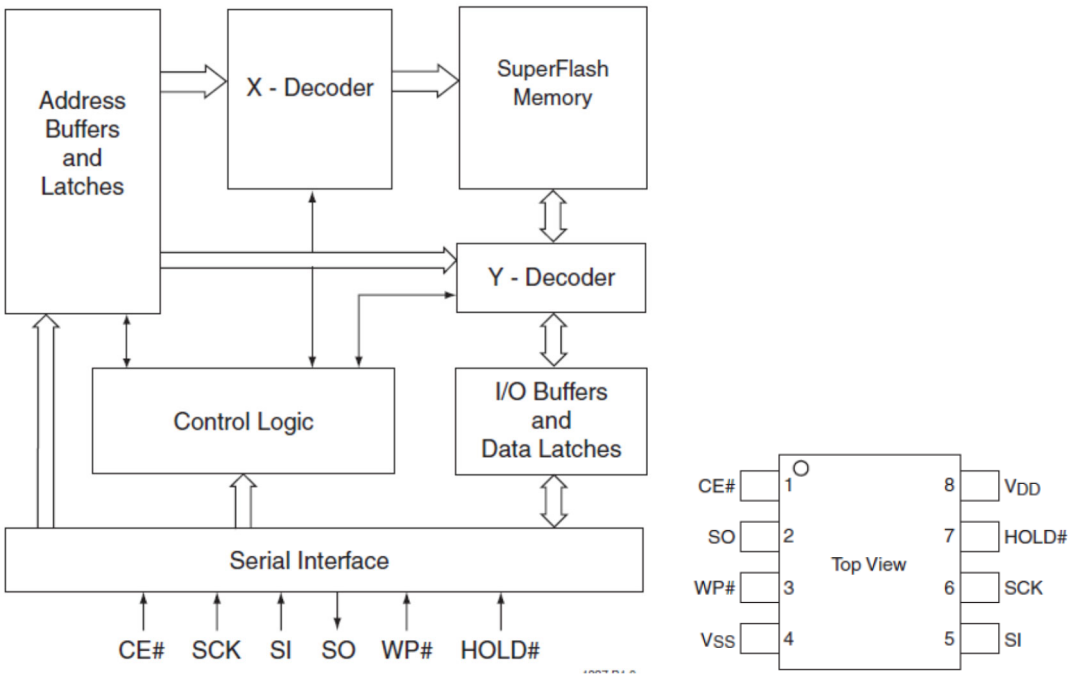
4. A matrix LED as shown in the figure below is to be interfaced to a microprocessor. Explain how the interfacing will be done and the factors that should be considered.



## Optional

5. The SPI is a common interface for a microprocessor to communicate with peripheral chips. In order to communicate effectively, the programmer must have a good understanding of the protocols associated with that particular chip. These protocols may be unique to that particular chip or may be generalized due to an industry standard.

   We will look at some of the protocols of the 32Mbit SPI Flash chip (SST25VF032B) to have a better understanding.

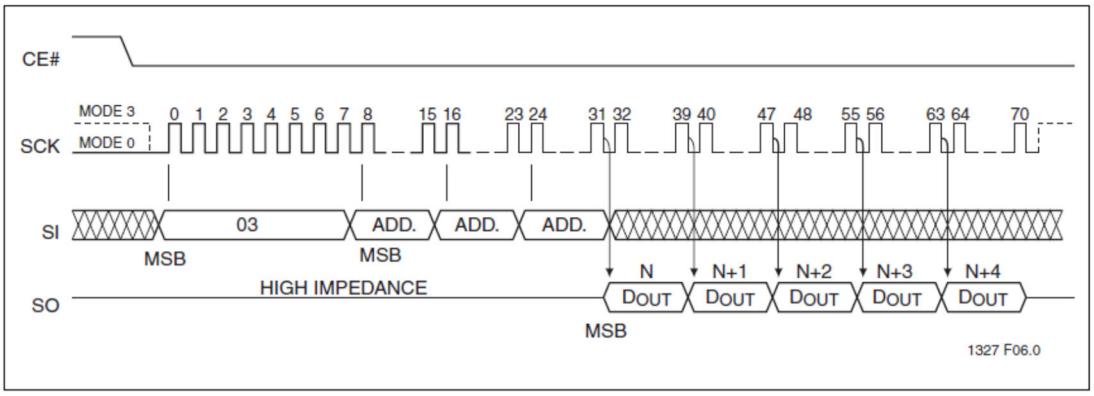



Block Diagram and Pin-Out of the SPI Flash Chip

The Device Operation Protocol is as follows:

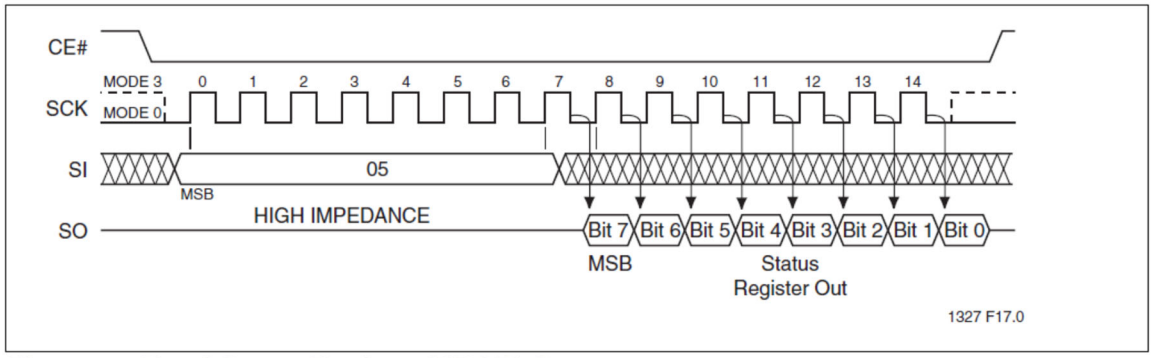
| Instruction | Description | Op Code Cycle[1] | Address Cycle(s)[2] | Dummy Cycle(s) | Data Cycle(s) | Maximum Frequency |
|---|---|---|---|---|---|---|
| Read | Read Memory | 0000 0011b (03H) | 3 | 0 | 1 to ∞ | 25 MHz |
| High-Speed Read | Read Memory at higher speed | 0000 1011b (0BH) | 3 | 1 | 1 to ∞ | 80 MHz |
| 4 KByte Sector-Erase[3] | Erase 4 KByte of memory array | 0010 0000b (20H) | 3 | 0 | 0 | 80 MHz |
| 32 KByte Block-Erase[4] | Erase 32KByte block of memory array | 0101 0010b (52H) | 3 | 0 | 0 | 80 MHz |
| 64 KByte Block-Erase[5] | Erase 64 KByte block of memory array | 1101 1000b (D8H) | 3 | 0 | 0 | 80 MHz |
| Chip-Erase | Erase Full Memory Array | 0110 0000b (60H) or 1100 0111b (C7H) | 0 | 0 | 0 | 80 MHz |
| Byte-Program | To Program One Data Byte | 0000 0010b (02H) | 3 | 0 | 1 | 80 MHz |
| AAI-Word-Program[6] | Auto Address Increment Programming | 1010 1101b (ADH) | 3 | 0 | 2 to ∞ | 80 MHz |
| RDSR[7] | Read-Status-Register | 0000 0101b (05H) | 0 | 0 | 1 to ∞ | 80 MHz |
| EWSR | Enable-Write-Status-Register | 0101 0000b (50H) | 0 | 0 | 0 | 80 MHz |
| WRSR | Write-Status-Register | 0000 0001b (01H) | 0 | 0 | 1 | 80 MHz |
| WREN | Write-Enable | 0000 0110b (06H) | 0 | 0 | 0 | 80 MHz |
| WRDI | Write-Disable | 0000 0100b (04H) | 0 | 0 | 0 | 80 MHz |
| RDID[8] | Read-ID | 1001 0000b (90H) or 1010 1011b (ABH) | 3 | 0 | 1 to ∞ | 80 MHz |
| JEDEC-ID | JEDEC ID read | 1001 1111b (9FH) | 0 | 0 | 3 to ∞ | 80 MHz |

The following figure shows the waveform for the 'Read Sequence' command.




a. Decode the protocol sequence from the microprocessor's perspective.

The following figure shows another SPI transaction with the device.




b. Decode the protocol command.

c. Decode the protocol sequence from the microprocessor's perspective.